MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

ICT DEPARTMENT

END OF SEMESTER EXAMINATIONS

BACHELOR OF SCIENCE (ICT)

ICT4702 MODELING AND SIMULATION

DATE: SEPTEMBER , 2017 TIME ALLOWED: 3HRS

Instructions:

Answer **ALL** four questions.

1.

(a) For each of the following systems, state ONE entity, TWO events and TWO state variables:

(i) Cafeteria **[5marks]**

(ii) Hospital Emergency Room **[5marks]**

(b) In a simulation study, a modeler generally adopts a world view or orientation for developing a model. Describe, any THREE world views you would adopt in developing a typical model. **[6marks]**

(c) List Processing is one of the key routines in a simulation study. Compare List Processing using arrays to List Processing using dynamic allocation of linked list **[8marks]**

(d) There are many features that are relevant when selecting simulation software. Briefly discuss the following broad classes of features

(i) Model-building features **[4marks]**

(ii) Runtime environment features **[4marks]**

(iii) Output features **[4marks]**

(iv) Animation and layout features **[4marks]**

2. Students arrive at the university library counter at random times that are from 1 to 8 minutes apart. Each possible value of the interarrival time has the same probability of occurrence. The time for a transaction at the counter is distributed as follows.

Transaction time (Minutes)	1	2	3	4	5	6
Probability	0.10	0.20	0.30	0.25	0.10	0.05

If two students are in the queue, an arriving student goes away without joining the queue. Simulate the system and determine the balking rate of the students. Use the following *(StudentIdentity, InterArrivalTimeRandomDigit, TransactionTimeRandomDigit)* triples list representing the first 15 students. Order the list by *StudentIdentity*.

(1, - , 84) ; (2, 064, 18); (3, 112, 87); (4, 678, 81); (5, 289, 06); (6, 871, 91); (7, 583, 79); (8, 139, 09); (9, 423, 64); (10, 039, 38); (11, 413, 94); (12, 462, 32); (13, 843, 79); (14, 738, 92); (15, 359, 46)

[20marks]

3. A random variable X that has a probability mass function given by $p(x)=\frac{1}{n+1}$ over the range $R_x=\{0,1,2,...,n\}$ is said to have a discrete uniform distribution.

(a) Using the series summations $\sum_{i=1}^{n} i=\frac{n(n+1)}{2}$ and $\sum_{i=1}^{n} i^2=\frac{n(n+1)(2n+1)}{2}$, find

(i) The mean of the distribution **[5 marks]**

(ii) The variance of the distribution **[5 marks]**

(b) If $R_x=\{a, a+1, a+2, ..., b\}$ compute

(i) The mean of the distribution **[5 marks]**

(ii) The variance of the distribution **[5 marks]**

4. For 16-bit computers, L'Ecuyer recommends combining three multiplicative generators, with $m_1 = 32363$, $a_1 = 157$, $m_2 = 31727$, $a_2 = 146$, $m_3 = 31657$ and $a_3 = 142$. The period of this generator is approximately 8×10^{12}. Generate 5 random numbers with the combined generator, using the initial seeds $X_{i,0} = 100,\ 300,\ 500,$ for the individual generators $i = 1,2,3$.

[20 marks]

END OF QUESTION PAPER